

Applied Industrial Organization

Classwork 4

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The database `verboven_cars.dta` contains information about car sales. In particular:

- `year`
 - `market` represent the geographical unit
 - `code_model` is the model of the car
 - `quantity` is the number of sales
 - `price`
 - `pop` is the market population (i.e. *market dimension*)
 - `li`, `wi`, `cy`, `we`, `pl`, `do`, `le`, `sp` are product's characteristics.
1. Create a Word file named `yourname_yoursurname_cw4.docx`
 2. Create a new do-file and save it as `yourname_yoursurname_cw4.do`
 3. Import the database into Stata.
 4. First, consider a single product model. Estimate the following regression:
 - (a) Raw price elasticity
 - (b) Add to the previous regression, model (`code_model`) fixed-effects
 - (c) Add to the previous regression, year fixed-effects
 - (d) Add to the previous regression, market fixed-effectsInclude the output in the Word file and comment on the results obtained.
 5. Add in the last regression a control for the population and include it in the Word file. Is the coefficient identified? Why?

6. Now, you want to estimate a model in *characteristics-space*. Estimate the appropriate regression and then add control functions following the same path applied in point 4 (4 different regressions). Include the output in the Word file and comment on the results obtained.
7. Now include product characteristics in the last regression. Include the output in the Word file and comment on the results obtained.
8. You suspect that the price elasticity may still be biased. So you decide to instrument price with the **BLP** instrument obtained from characteristics **sp**.
Hint: The instrument needs to be computed *within* (by) market and year and *across* products.
9. Now, you want to estimate the *Costant Elasticity of Substitution* (CES) Demand System for two different car models: Renault 4 (`code_model == 196`) and Audi 80/90 (`code_model == 13`) using as a reference product (price and quantity **1**) the Rover Mini (`code_model == 217`). Obtain the parameter σ for each of the two products. Include in a Word file and comment on the results. Does the model seem appropriate?
Hint:
 - Keep only the three products listed above
 - Keep only the variables that you need
 - Then, you need to *reshape* the data.